



FIELDS

Mathematics Education Forum

March 28, 10 AM – 2 PM

Fields Institute, 222 College Street, Toronto

Tweet us @FieldsMathEd

Reframing the Impact of Assessment in MathEd, from Measurement to Meaning

Frosina Stojanovska-Pocuca (Mohawk College) & Charles Anifowose (Vretta Inc.)

Assessment data plays an important role in shaping mathematics education, influencing classroom practice and curriculum design. As Ontario reflects on recent large-scale assessment results and evolving approaches to measuring student achievement, this forum brings together researchers, administrators, and teachers to examine how assessment data can deepen understanding of student learning and support meaningful action. Drawing on examples from international assessments such as PISA (Programme for International Student Assessment) and provincial data from EQAO (Education Quality and Accountability Office), participants will explore what assessment results reveal about mathematical literacy, student achievement pathways, and readiness for post-secondary study. The discussion will focus on how educators can move beyond reporting test scores and instead use assessment evidence to identify learning gaps, strengthen preparation, and inform teaching and policy decisions.

10:00 AM – 10:05 AM

Welcome to the Forum

Description: Online participants may join the waiting room prior to the start of the Forum. All forums are preceded by a land acknowledgement, an introduction to the day's sessions, and reports from our community of mathematics education organizations (AFEMO, CMESG, CMS, CME, FYMSiC, OAME, OMCA, OCMA, OCMC, and others).

10:05 AM – 10:45 AM

PISA, Math Literacy, and Student Outcomes Across the Industrialized World: An Ordinary Least Squares Analysis with Policy Implications

Louis Volante (Brock University)

This talk discusses findings from a recent Ordinary Least Squares (OLS) analysis of the impact of the pandemic on cognitive and non-cognitive learning outcomes across the industrialized world. More specifically, mathematical literacy results from the 2022 Programme for International Student Assessment (PISA) were juxtaposed against five previous test administrations to examine the impact of school closures across Canada, United States, Australia, New Zealand, Great Britain, and Europe. The findings, gleaned from approximately 1.45 million student test profiles, suggested an important intersection between cognitive and non-cognitive outcomes. In particular, pandemic-related learning deficits for students with high levels of a sense of belonging and growth mindset were approximately half a school year for mathematics. Conversely, for students who were frequently bullied, the COVID-related losses averaged 1.5 school years. Overall, the analysis indicated that students, when controlled for the trend, experienced a loss of 0.07 standard deviations in their sense of belonging to school, an increase of 0.14 standard deviations in fixed mindset, and a 3% decrease in the probability of being frequently bullied, compared to past student cohorts. The seminar concludes with a discussion of the long-term implications of the current findings for mathematics literacy in both compulsory and higher education systems.

Louis Volante (PhD) is a Distinguished Professor at Brock University and a Professorial Fellow at UNU-MERIT / Maastricht Graduate School of Governance. His interdisciplinary scholarship focuses on the global governance of education; politics, policy, and large-scale reform; political economy of education systems; international achievement surveys and policy diffusion; and impact evaluation of policies and programmes. Professor Volante's research is widely referenced in academic and policy communities and has received continuous funding from the Social Sciences and Humanities Research Council of Canada (SSHRC). His current SSHRC-funded research examines the long-term impact of the pandemic on student learning outcomes.

10:45 AM – 11:05 AM

Assessment for all – addressing assessment issues in mathematics education.

Guri A. Nortvedt (University of Oslo)

The main aim of assessment in mathematics education is to provide evidence that can be used to make decisions about or improve mathematics education, raising questions about which aspects of students' mathematical competence should be assessed—as well as how and for what purpose. This lecture will address 1) issues related to the assessment process and to the development of assessments that can validly assess students' mathematical competence; (2) issues related to educational policy and policy-making based on assessment data; and (3) issues related to equity. A main aim of the presentation is to show how these issues play out differently for different stakeholders, for instance students, teachers, school leaders and educational authorities.

Professor Guri A. Nortvedt works at the University of Oslo, Norway. Her research focuses on various aspects of educational assessment in mathematics education, including both classroom and large-scale external assessments. Professor Nortvedt's work encompasses themes such as culturally responsive assessment, teacher assessment literacy, formative assessment, and assessment of diverse students.

11:05 AM – 11:45 AM

Tracking Pathways in Mathematics Achievement: EQAO Cohort Data for Classroom Reflection

Maggie Dunlop (EQAO), **Elli Weisdorf** (EQAO)

How are students faring in mathematics as they progress through school in Ontario? Which groups of students are gaining ground, and which students are falling behind? Are trends the same as in the past? Presenters will share findings from a project in which EQAO linked the 2021–2022 and 2024–2025 assessment outcomes in mathematics of students who participated in the primary (Grade 3), junior (Grade 6) and Grade 9 assessments. We explore instructional implications across grades and for key student groups, and offer insights to inform future teaching and learning.

At EQAO, Maggie works primarily on projects that focus on identifying instructional insights for use by Ontario educators. She completed her doctoral studies at OISE with Dr. Eunice Jang, specializing in educational measurement. Maggie now has around 20 years' experience across the Northern Hemisphere in program evaluation and research initiatives that have mainly aimed to improve language, literacy and mathematics learning outcomes for a range of learners. Maggie uses a range of quantitative and qualitative methodologies in her work. Her interests lie in technology in education, creativity,

outdoor education, educational program evaluation, language and literacy learning, and mathematics learning.

Elli Weisdorf is an elementary educator and OISE Doctor of Education student with over 25 years of professional experience and a deep commitment to equitable, research-informed mathematics instruction. As a seconded Education Officer for School Support & Outreach at EQAO, she works with school and board teams to interpret assessment data and inform improvement planning. A former Elementary Mathematics Facilitator and current AQ Instructor with the York Region District School Board, Elli is also an author and advocate dedicated to removing barriers to student success through inclusive, anti-oppressive math practices.

11:45 AM – 12:00 PM

Mining Insights from an Assessment Task

Robin Sharma (McGill University, Learning Design Collab)

This session explores how assessment of students' mathematical problem solving can be used to generate evidence about their thinking. Rather than focusing only on scores, the talk examines how structural analysis informed by a relevant theory of mathematical cognition can produce formative insights. An example task will be shown to demonstrate how structural analysis (leveraging digital delivery environments and LLMs) can be used to unravel the reasoning, strategies, and misconceptions in student responses. Participants will see how different responses can reveal the range of students' understanding and misconceptions.

Robin is a learning scientist who creates science and theory-based Digital Learning Environments (DLEs) for teachers and children. Then, he studies the effects of such DLEs on academic achievement, attitudes, and acceptance. He uses experimental designs to generate evidence and make recommendations for policy, implementation, and curricular integration of educational technologies. Specifically, his research has focused on the meaningful design and evaluation of digital games for building geometrical understanding of adolescents. He applies theories of mathematical cognition, educational game design, and emotional design principles to create math games.

12:00 PM – 1:00 PM

LUNCH BREAK

(Light refreshments provided)

Online participants will have an extra hour to work through the workshop tasks during this period.

1:00 PM – 2:00 PM

Breakout Workshop: Tasks, Responses, and Cognitive Models

This workshop examines how assessment tasks function in practice. Participants will work in small breakout groups with a set of 2 to 4 mathematical tasks ([Link to Tasks](#)), using them as a starting point for analyzing how tasks produce evidence about student thinking. For each task, groups will consider:

- What mathematical ideas does the task target?
- What strategies might students use to solve it?
- What misconceptions might appear in student responses?
- Does the task effectively measure the intended mathematical understanding?
- What features of the task support or hinder student reasoning?
- How could the task be adapted to better reveal student thinking?

The workshop concludes with the construction of a shared cognitive model derived from the breakout discussions. This model is not intended to serve as a full psychometric framework. Instead, it acts as a practical map of the thinking, strategies, misconceptions, representations, and inferential limits associated with the selected tasks. We will then discuss the use of this model as a practical tool for understanding what student responses reveal, and do not reveal, about mathematical thinking.

2:00 PM ADJOURNMENT